

Tier R — Remediate: Using a Table to Evaluate Inverse Trig — Key

1. $\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

2. $\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$

3. $\arctan(\sqrt{3}) = \frac{\pi}{3}$

4. $\arcsin(-1) = -\frac{\pi}{2}$

5. $\arccos(-1) = \pi$

6. $\arctan(0) = 0$

Tier A — Apply: Domain, Range, and Compositions — Key

7.

Function	Domain	Range
$f(x) = \arcsin x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$g(x) = \arccos x$	$[-1, 1]$	$[0, \pi]$
$h(x) = \arctan x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

8. $\sin(\arccos \frac{3}{5})$

Let $\theta = \arccos \frac{3}{5}$, so $\cos \theta = \frac{3}{5}$. Right triangle with adjacent = 3, hypotenuse = 5, opposite = $\sqrt{25 - 9} = 4$. Therefore $\sin \theta = \frac{4}{5}$.

9. $\tan(\arcsin \frac{5}{13})$

Let $\theta = \arcsin \frac{5}{13}$, so $\sin \theta = \frac{5}{13}$. Opposite = 5, hypotenuse = 13, adjacent = $\sqrt{169 - 25} = 12$. Therefore $\tan \theta = \frac{5}{12}$.

Tier E — Extend: Compositions and Domain Analysis — Key

10.

(a) $\sin(\arcsin \frac{1}{2}) = \frac{1}{2}$

(b) $\arcsin(\sin \frac{\pi}{3}) = \frac{\pi}{3}$

(c) $\arcsin(\sin \frac{2\pi}{3}) = \frac{\pi}{3}$ (since $\frac{2\pi}{3} \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$, use (d) $\cos(\arccos(-0.7)) = -0.7$

$\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, then $\arcsin(\frac{\sqrt{3}}{2}) = \frac{\pi}{3}$

11. $\sin(\arcsin x) = x$ for all $x \in [-1, 1]$, because $[-1, 1]$ is the domain of $\arcsin$.

12. $\arcsin(\sin x) = x$ only for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. For other values, $\sin x$ equals the same value at multiple angles, and $\arcsin$ always returns the principal value in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, not necessarily x itself.

13. Let $\theta = \arctan x$ with $x > 0$, so $\tan \theta = x$. Right triangle: opposite = x , adjacent = 1, hypotenuse = $\sqrt{x^2 + 1}$.

$$\cos(\arctan x) = \frac{1}{\sqrt{x^2 + 1}}$$